GATE ME 90 Days Challenge

Day 67/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Engineering Mathematics	**Maths Revision**	**33 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on a revision of all of Engineering Mathematics: eigenvalues and special matrices, limits, maxima and integration, curl and Green's theorem, first- and second-order ODEs, analytic functions and residues, Newton–Raphson and Gauss–Seidel, and probability distributions. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE. **Today's test:** open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

For the matrix A with rows (4, −2) and (1, 1), the largest eigenvalue of $A^3 - 2A$ is _____

Q2 MCQ

A is a real skew-symmetric matrix of order 5. Then

(A) det A = 1
(B) det A = 0
(C) A is always invertible
(D) all eigenvalues of A are real and non-zero

Q3 NAT

The value of the limit of $(\cos x)^{1/x^2}$ as $x \to 0$ (round off to three decimal places) is _____

Q4 MCQ

The maximum value of $f(x) = x\, e^{-x}$ for $x \geq 0$ is

(A) 1
(B) e
(C) 1/e
(D) $2/e^2$

Q5 NAT

The value of the integral of $x \ln x$ from $x = 0$ to $x = 1$ is _____

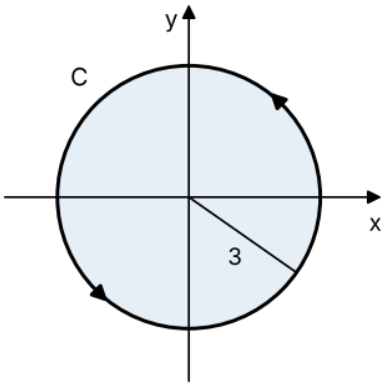
Q6 MCQ

The curl of the vector field $F = yz\,i + zx\,j + xy\,k$ is

(A) $x\,i + y\,j + z\,k$
(B) $2(x\,i + y\,j + z\,k)$
(C) $i + j + k$
(D) 0

Q7 NAT

Using Green's theorem, the value of the line integral of $(x^2 - y)\,dx + (x + y^2)\,dy$ taken anticlockwise around the circle C: $x^2 + y^2 = 9$ shown (round off to two decimal places) is _____



Q8 MCQ

The solution of $dy/dx = xy^2$ with $y(0) = 1$, evaluated at $x = 1$, is

(A) 2
(B) 1/2
(C) $e^{1/2}$
(D) 3/2

Q9 MCQ

The order and degree of the differential equation $(d^2y/dx^2)^3 + (dy/dx)^4 + y = 0$ are

(A) order 3, degree 2
(B) order 2, degree 4
(C) order 2, degree 3
(D) order 4, degree 3

Q10 NAT

C is the circle $|z - 1| = 2$ traversed anticlockwise. The value of $(1/2\pi i)$ times the integral of $(z + 2)/[z(z - 4)]$ around C is _____

Q11 MCQ

$u = e^x \cos y$ is the real part of an analytic function f(z). A harmonic conjugate v is

(A) $e^x \cos y$
(B) $e^x \sin y$
(C) $-e^x \sin y$
(D) $e^{-x} \sin y$

Q12 NAT

$\sqrt{10}$ is computed by the Newton–Raphson method applied to $f(x) = x^2 - 10$, starting from $x_0 = 3$. The first iterate x_1 (round off to three decimal places) is _____

Q13 MCQ

The Gauss–Seidel iteration for $Ax = b$ is guaranteed to converge for any starting vector when A is

(A) strictly diagonally dominant
(B) singular
(C) skew-symmetric
(D) upper triangular with zero diagonal

Q14 NAT

Two fair dice are thrown and S is the sum of the numbers shown. The variance of S (round off to two decimal places) is _____

Q15 MCQ

The median of an exponential distribution with rate parameter λ (mean 1/λ) is

(A) 1/λ
(B) λ ln 2
(C) 1/(2λ)
(D) (ln 2)/λ

Answer Key

Q	Type	Answer
Q1	NAT	21
Q2	MCQ	(B) det A = 0
Q3	NAT	0.607 (accept 0.601 to 0.613)
Q4	MCQ	(C) 1/e
Q5	NAT	-0.25
Q6	MCQ	(D) 0
Q7	NAT	56.55 (accept 56.0 to 57.1)
Q8	MCQ	(A) 2
Q9	MCQ	(C) order 2, degree 3
Q10	NAT	-0.5
Q11	MCQ	(B) $e^x \sin y$
Q12	NAT	3.167 (accept 3.16 to 3.17)
Q13	MCQ	(A) strictly diagonally dominant
Q14	NAT	5.83 (accept 5.80 to 5.87)
Q15	MCQ	(D) (ln 2)/λ

Detailed Solutions

Q1 **Answer: 21**

Concept: If $Ax = \lambda x$, then $p(A)x = p(\lambda)x$ for any polynomial p: the eigenvalues of p(A) are $p(\lambda)$.

Characteristic equation $\lambda^2 - 5\lambda + 6 = 0$ (trace 5, determinant 4 + 2 = 6) → $\lambda = 2, 3$

$p(\lambda) = \lambda^3 - 2\lambda$: $p(2) = 8 - 4 = 4$; $p(3) = 27 - 6 = 21$

Largest eigenvalue = **21**

Exam tip: Never compute A^3 explicitly in the exam; work with the eigenvalues.

Q2 **Answer: (B) det A = 0**

Concept: $A^T = -A$, so $\det A = \det A^T = \det(-A) = (-1)^n \det A$.

For odd n = 5: $\det A = -\det A$, hence **det A = 0 (B)**

The eigenvalues of a real skew-symmetric matrix are zero or purely imaginary (±iβ); with odd order, one eigenvalue must be 0.

Exam tip: The diagonal entries of a skew-symmetric matrix are all zero, so its trace is 0.

Q3 **Answer: 0.607 (accept 0.601 to 0.613)**

Concept: A 1^∞ form: $\lim f^g = e^{\lim g(f-1)}$ (or take logarithms and use L'Hospital's rule).

$\ln L = \lim (\ln \cos x)/x^2$; $\ln \cos x \approx -x^2/2$ for small x

$\ln L = -1/2$, so $L = e^{-1/2} = \mathbf{0.607}$

Exam tip: Memorise $\cos x \approx 1 - x^2/2$ and $\ln(1 + u) \approx u$; most 1^∞ limits fall in one line.

Q4 **Answer: (C) 1/e**

Concept: Stationary point from $f'(x) = 0$, confirmed by the sign of f''.

$f'(x) = e^{-x}(1 - x) = 0 \rightarrow x = 1$; $f''(1) = e^{-1}(1 - 2) < 0$, a maximum

$f(1) =$ **1/e (C)** ≈ 0.368

Exam tip: $2/e^2$ is $f(2)$, the point of inflection, a common distractor.

Q5 **Answer: −0.25**

Concept: Integration by parts with $u = \ln x$, $dv = x\,dx$; the lower limit uses $x^2 \ln x \rightarrow 0$ as $x \rightarrow 0$.

$\int x \ln x\,dx = (x^2/2)\ln x - \int (x^2/2)(1/x)\,dx = (x^2/2)\ln x - x^2/4$

From 0 to 1: $(0 - 1/4) - (0 - 0) =$ **−0.25**

Exam tip: In general, $\int_0^1 x^n \ln x\,dx = -1/(n + 1)^2$.

Q6 **Answer: (D) 0**

Concept: $\nabla \times F = (\partial F_z/\partial y - \partial F_y/\partial z)\,i + (\partial F_x/\partial z - \partial F_z/\partial x)\,j + (\partial F_y/\partial x - \partial F_x/\partial y)\,k$.

i: $x - x = 0$; j: $y - y = 0$; k: $z - z = 0 \rightarrow$ **0 (D)**

F is conservative: $F = \nabla(xyz)$.

Exam tip: Spot a potential first; if $F = \nabla\varphi$, the curl is zero and line integrals depend only on the end points.

Q7 **Answer: 56.55 (accept 56.0 to 57.1)**

Concept: Green's theorem: $\oint(P\,dx + Q\,dy) = \iint(\partial Q/\partial x - \partial P/\partial y)\,dA$ over the enclosed region.

$P = x^2 - y$, $Q = x + y^2$; $\partial Q/\partial x - \partial P/\partial y = 1 - (-1) = 2$

Integral $= 2 \times$ area $= 2 \times 9\pi = 18\pi =$ **56.55**

Exam tip: The x^2 and y^2 terms contribute nothing; only the 'cross' terms survive.

Q8 **Answer: (A) 2**

Concept: Variables separable: $dy/y^2 = x\,dx$.

$-1/y = x^2/2 + C$; $y(0) = 1$ gives $C = -1$, so $y = 1/(1 - x^2/2) = 2/(2 - x^2)$

$y(1) = 2/(2 - 1) =$ **2 (A)**

Exam tip: This solution blows up at $x = \sqrt{2}$: non-linear ODEs can have finite-escape times.

Q9 **Answer: (C) order 2, degree 3**

Concept: Order = highest derivative present; degree = power of that highest derivative when the equation is polynomial in its derivatives.

Highest derivative $d^2y/dx^2 \rightarrow$ order 2; it appears cubed $\rightarrow$ degree 3: **(C)**

Exam tip: The power of a lower derivative (here 4 on dy/dx) does not decide the degree.

Q10 **Answer: −0.5**

Concept: Residue theorem: $(1/2\pi i)\oint f\,dz =$ sum of residues at the poles inside C.

Poles: $z = 0$ (distance 1 from the centre 1, inside) and $z = 4$ (distance 3, outside)

Residue at $0 = (0 + 2)/(0 - 4) = -0.5 \rightarrow$ answer **−0.5**

Exam tip: Always check each pole's distance from the centre of the circle, not from the origin.

Q11 **Answer: (B) $e^x \sin y$**

Concept: Cauchy–Riemann: $v_y = u_x$ and $v_x = -u_y$.

$v_y = e^x \cos y \rightarrow v = e^x \sin y + g(x)$; $v_x = e^x \sin y = -u_y$ ✓, so g is constant

$v = \mathbf{e^x \sin y}$ **(B)**, and $f(z) = e^z$

Exam tip: Milne–Thomson: put $x = z$, $y = 0$ in $f' = u_x - iu_y$ to get $f'(z) = e^z$ directly.

Q12 **Answer: 3.167 (accept 3.16 to 3.17)**

Concept: $x_{n+1} = x_n - f(x_n)/f'(x_n)$; for square roots this is $x_{n+1} = (x_n + 10/x_n)/2$.

$x_1 = 3 - (9 - 10)/(2 \times 3) = 3 + 1/6$

$x_1 = \mathbf{3.167}$ (true $\sqrt{10} = 3.1623$)

Exam tip: Convergence is quadratic: $x_2 = 3.16228$ already has 5 correct digits.

Q13 **Answer: (A) strictly diagonally dominant**

Concept: A sufficient condition for Jacobi and Gauss–Seidel convergence is strict diagonal dominance: $|a_{ii}| > \Sigma_{j \neq i} |a_{ij}|$ in every row.

Hence **(A)**; Gauss–Seidel also converges for symmetric positive-definite matrices.

Exam tip: If the given system is not diagonally dominant, reorder the equations before iterating.

Q14 **Answer: 5.83 (accept 5.80 to 5.87)**

Concept: For independent variables, variances add: Var(X + Y) = Var X + Var Y.

One die: $E[X] = 3.5$, $E[X^2] = 91/6$, $\text{Var } X = 91/6 - 12.25 = 35/12$

$\text{Var } S = 2 \times 35/12 = 35/6 = \mathbf{5.83}$

Exam tip: For a discrete uniform on 1…n, $\text{Var} = (n^2 - 1)/12$; n = 6 gives 35/12.

Q15 **Answer: (D) (ln 2)/λ**

Concept: The median m satisfies $F(m) = 1 - e^{-\lambda m} = 1/2$.

$e^{-\lambda m} = 1/2 \rightarrow m =$ **(ln 2)/λ (D)** $\approx 0.693/\lambda$

Since the distribution is right-skewed, the median is less than the mean $1/\lambda$, and the mode is 0.

Exam tip: This is the 'half-life' formula of radioactive decay.

What next?

Day 67 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

Dig Career Thrust app

Telegram: daily PDFs

Next, Day 68/90: General Aptitude, GA Revision. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.